GEOGRAPHY

PHYSICAL GEOGRAPHY

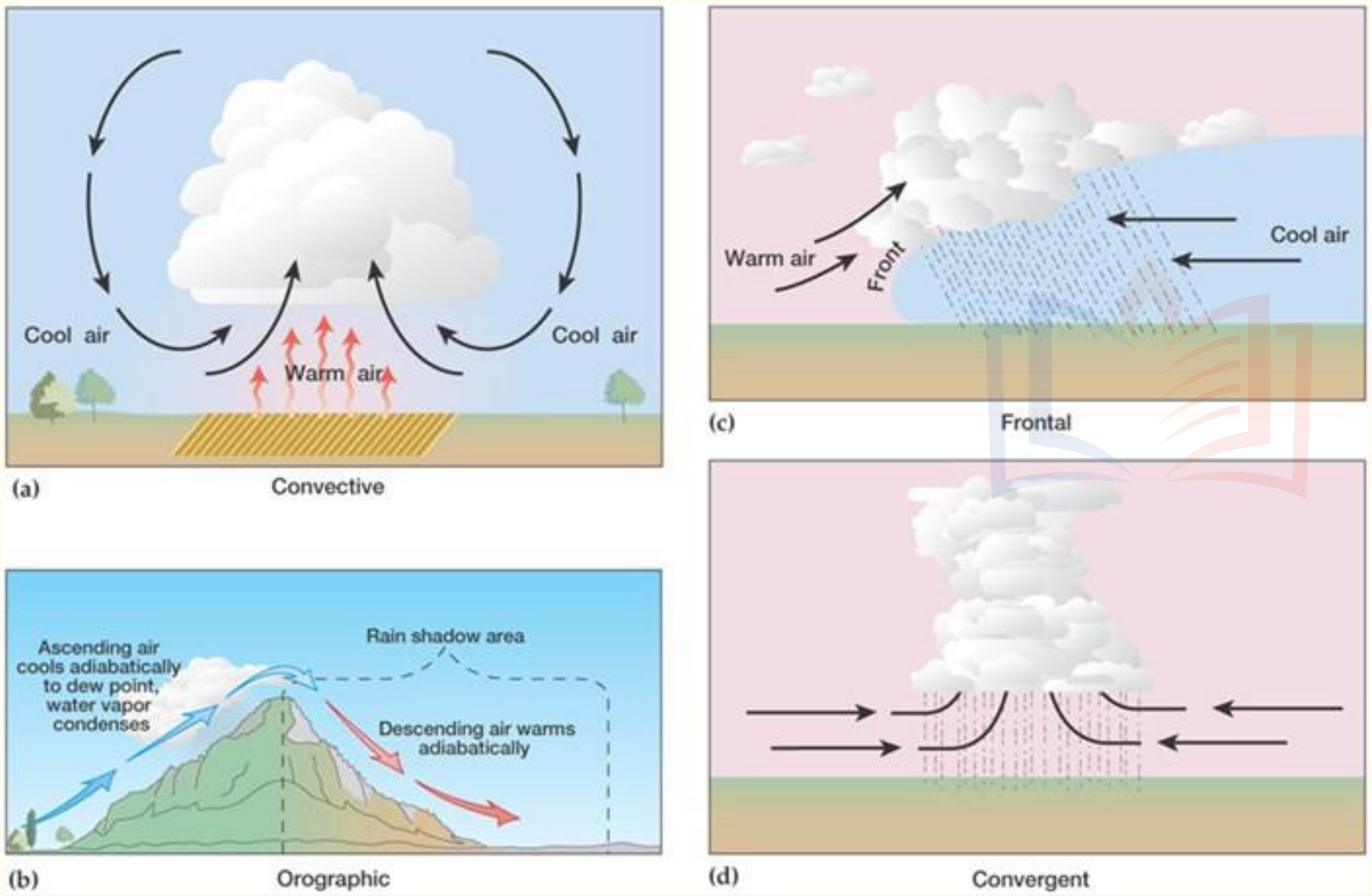
CLIMATOLOGY

MODULE – 29

- TYPES OF PRECIPITATION-2
- GLOBAL DISTRIBUTION OF PRECIPITATION

PRECIPITATION

TYPES OF PRECIPITATION



1 Although all forms of precipitation occur due to condensation of moisture in the air due to cooling of air through its uplift, the mechanism of uplift may vary from one instance to the other.

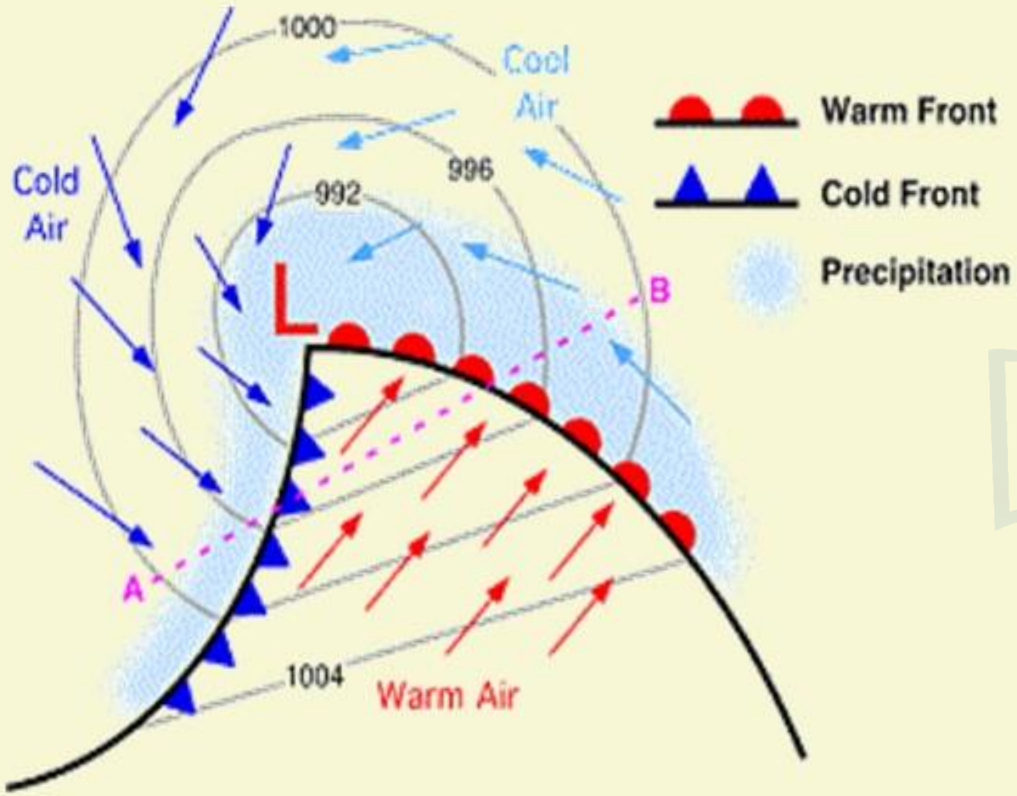
2 On the basis of the mechanism of uplift of air the precipitation can be

1. Convectional,
2. Orographic,
3. Cyclonic or Frontal.

PRECIPITATION

TYPES OF PRECIPITATION

3 CYCLONIC PRECIPITATION

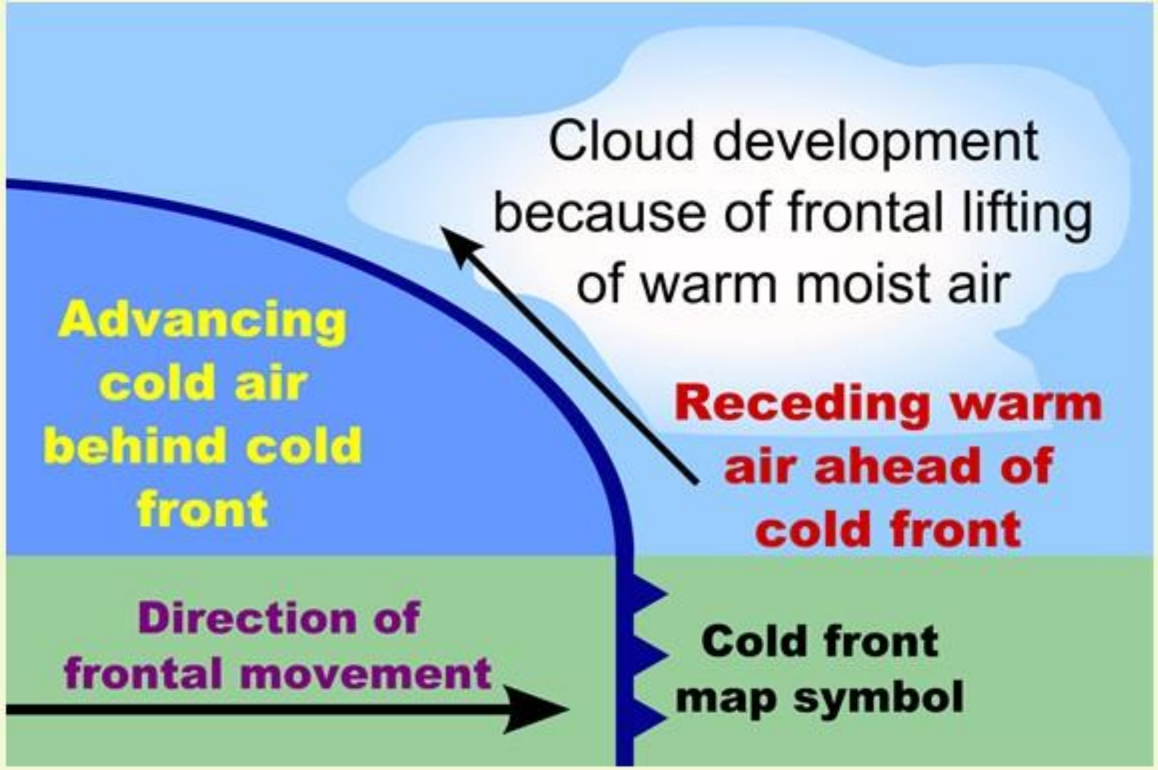


1. When the air is caused to rise upwards due to cyclonic circulation, the resulting precipitation is said to be one of the cyclonic type.
2. A cyclone is an area of low pressure and air from all around flows towards this central low pressure area.
3. The convergent flow of air leads to uplift of air and it causes heavy precipitation.

PRECIPITATION

TYPES OF PRECIPITATION

3 CYCLONIC PRECIPITATION



4 Cyclonic precipitation occurs in both tropical and temperate regions. Tropical cyclones cause heavy rainfall accompanied by high velocity winds.

5 Parts of Andhra Pradesh, Orissa and West Bengal are highly prone to such cyclones

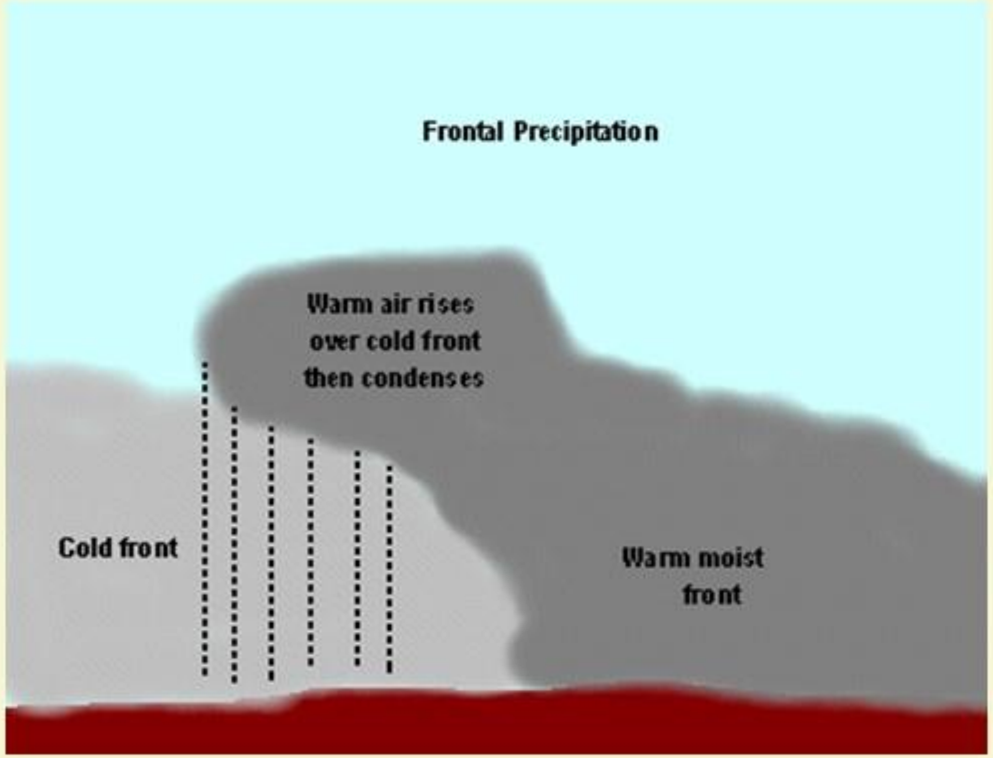
6 In contrast to the tropical cyclones the temperate cyclones are less intense. However, they also cause plenty of precipitation.

7 The winter rains in northern parts of India are caused by temperate cyclones called westerly disturbance in India.

PRECIPITATION

TYPES OF PRECIPITATION

3 CYCLONIC PRECIPITATION



8 The uplift of air caused due to formation of a front in an area where a cold air mass converges against a warm air mass is another form of cyclonic precipitation.

9 In such a situation the cold air lifts the warm air and it may result in condensation and precipitation.

10 Development of such fronts often leads to formation of temperature cyclones.

11 Hence such form of cyclonic precipitation is also called as frontal precipitation.

PRECIPITATION

GLOBAL DISTRIBUTION OF RAINFALL

1. Different places on the earth's surface receive different amounts of rainfall in a year and that too in different seasons.

2. In general, as we proceed from the equator towards the poles, rainfall goes on decreasing steadily.

3. The coastal areas of the world receive greater amounts of rainfall than the interior of the continents.

4. The rainfall is more over the oceans than on the landmasses of the world because of being great sources of water.

5. In the belt of trades, the rainfall is much heavier on the eastern side of the continent and decreases as it moves towards west. On the other hand, in zones where the westerlies have a strong influence the rainfall is first received on the western margins of the continents and it goes on decreasing towards the east.

6. Wherever mountains run parallel to the coast, the rain is greater on the coastal plain on the windward side and it decreases towards the leeward side.

PRECIPITATION

MOCK QUESTIONS

Q 9 Give a detailed account of the world distribution of precipitation.

PRECIPITATION

GLOBAL DISTRIBUTION OF RAINFALL

On the basis of the total amount of annual precipitation, major precipitation regimes of the world are identified as follows:

EQUATORIAL ZONE OF MAXIMUM RAINFALL

1. This zone extends up to 10^0 latitudes on either side of the equator and falls within intertropical convergence characterized by warm and moist air masses.
2. The mean annual rainfall ranges between 175cm and 200cm.
3. Most of the rains are received through convectional rainfall accompanied by lightening and cloud thunder.

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PRECIPITATION

GLOBAL DISTRIBUTION OF RAINFALL

EQUATORIAL ZONE OF MAXIMUM RAINFALL

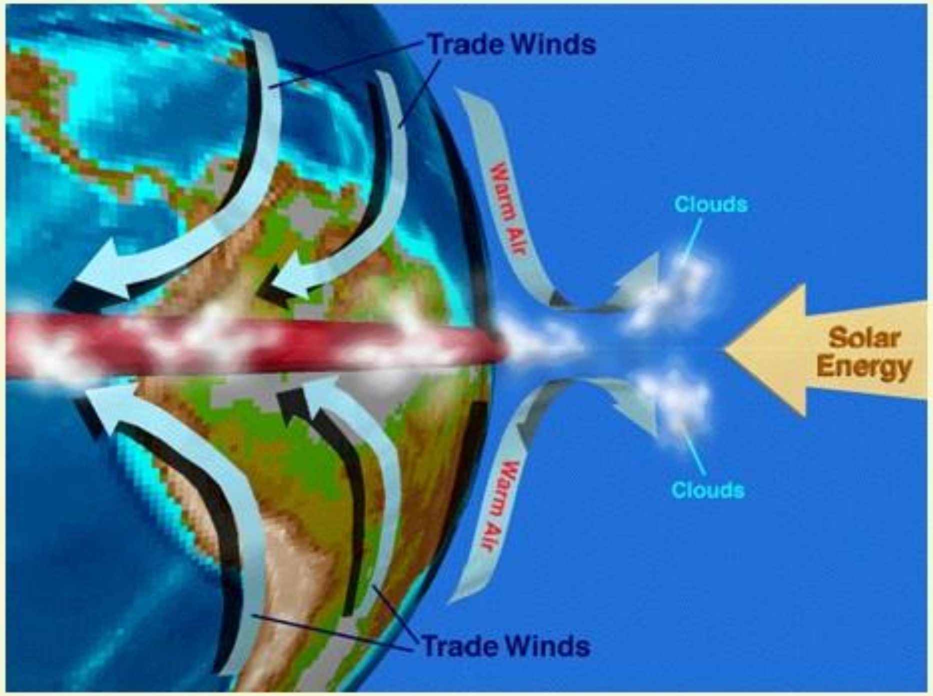
4 There is daily rainfall in the afternoon. The rainfall intensity is very high as it occurs in the form of heavy showers.

5 The clouds are cleared within short period and sky becomes cloudless in the late evening.

PRECIPITATION

GLOBAL DISTRIBUTION OF RAINFALL

TRADE WIND RAINFALL ZONE



1. Trade wind rainfall zone extends between 10^0-20^0 latitudes in both the hemispheres and is characterised by north-east and south-east trade winds.

2. These winds yield rainfall in the eastern parts of the continents because they come from over the oceans and hence pickup sufficient moisture but as they move westward in the continents they become dry and thus the western parts of the continents become extremely dry and deserts.

3. The monsoon regions located in this zone receive much rainfall. Summers receive most of the mean annual rainfall.

PRECIPITATION

GLOBAL DISTRIBUTION OF RAINFALL

SUBTROPICAL ZONE OF MINIMUM RAINFALL

1. Subtropical zone of minimum rainfall extends between 20^0-30^0 latitudes latitudes in both the hemispheres, where descending air from above induces high pressure and winds diverge in opposite directions at the ground surface, with the result anti-cyclones are formed.

2. This condition is not conducive for rainfall and hence dry conditions prevail over large areas.

3. Mean annual rainfall is 90 cm. It may be pointed out that all the tropical hot deserts are located in this zone where mean annual rainfall is below 25 cm.

4. The average annual rainfall for the whole zone is higher than the average value for the deserts because the eastern parts of the continents receive more rainfall from relatively moist trade winds which comes from over the oceans.

5. Most of annual rainfall occurs during summer months while winter season is dry.

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GLOBAL DISTRIBUTION OF RAINFALL

MEDITERRANEAN RAINFALL ZONE

Mediterranean rainfall zone extends between 30^0-40^0 latitudes in both the hemispheres where rainfall occurs through westerlies and cyclones during winter season while summers remain dry because this zone comes under the influence of trade winds due to northward shifting of wind and pressure belt during northern summer (summer solstice). Mean annual rainfall is 100 cm.

PRECIPITATION

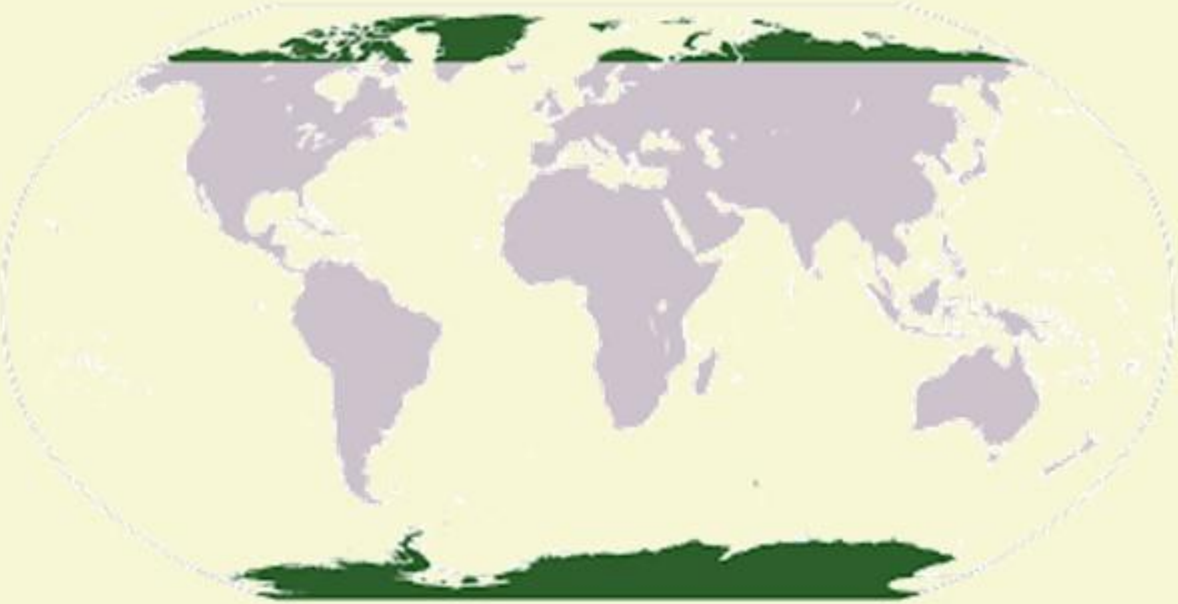
GLOBAL DISTRIBUTION OF RAINFALL

MID-LATITUDINAL ZONE OF HIGH RAINFALL

1. Mid-latitudinal zone of high rainfall extends between 40°-50° latitudes in both the hemispheres where rainfall occurs through westerlies and temperate cyclones.
2. Mean annual rainfall ranges between 100-125cm.
3. The western parts of the continents receives more rainfall. It decreases from the western coastal areas inland.
4. Southern hemisphere records more rainfall than northern hemisphere because of dominance of oceans in the former.
5. Winter season receives maximum precipitation through temperate cyclones.

PRECIPITATION

GLOBAL DISTRIBUTION OF RAINFALL



POLAR ZONE OF LOW PRECIPITATION

1. Precipitation decreases from 60^0 latitude poleward in both the hemispheres.
2. Mean annual precipitation becomes only 25cm beyond 75^0 latitude. Most of the precipitation occurs in the form of snowfall.

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